Computed but not consulted: a self-audit of a generative drug-discovery campaign against KRAS in pancreatic cancer

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Abstract

We report a self-audit of a generative small-molecule campaign against KRAS in pancreatic ductal adenocarcinoma, conducted by the authors of the generative system. It produced 73 candidate molecules; three were sent to synthetic chemists and approximately one year of laboratory work produced no compound. The failure was not one of prediction. The campaign's own output file contains a column named "Epoxide Ring Present", set True for all three molecules sent for synthesis. The same file records predicted mutagenicity at a median of 0.93 across all 73 molecules and synthetic accessibility at 5.47 against 4.21 and 3.84 for the two reference drugs. Every signal required to halt the campaign was computed, written down, and shipped, and none was connected to a decision. Selection used docking thresholds alone, and the one criterion with authority over the outcome, "Good Docking Quality Overall", reads True for 72 of 73 molecules. Against a control of 240,214 molecules from eight published generative models, this is not a general property of generative chemistry: those models pass an unmodified published filter at 81 to 89 percent, against 6.8 percent here. The distinguishing feature was not the generator but the absence of any gate between prediction and purchase order. We rebuild the pipeline so that every stage must reproduce known answers before its output is used, and report four validated targets, three completed screens, and two occasions on which our own replacement methods failed in the same manner and were caught by those gates. We also report four hypotheses we proposed for the failure and could not support, including our own, and state where the surviving claim is bounded: the audit is of a single campaign, the laboratory outcome is attested rather than documented, and no compound has been measured. Code, data, and all negative results are released.

1. Introduction

Computational drug-discovery pipelines are assembled from components that are individually validated and collectively unaudited. A generative model proposes molecules, a docking method scores them, property predictors annotate them, and a shortlist emerges. Each component has a literature establishing what it does well. Very little establishes what happens when they are chained and the chain is trusted.

The limitations of the individual components are known. Generative models propose molecules that cannot readily be synthesised (ref. 1). Structural-alert catalogues over-reject, and Capuzzi et al. report that **87 small-molecule FDA-approved drugs carry PAINS alerts**, cautioning explicitly against using such alerts to triage compounds (refs. 2, 3). Deep-learning docking methods produce physically invalid poses (ref. 4). The field has also been reviewed recently and comprehensively (ref. 14), and that review is accompanied by a curated compilation of 73 generative campaigns that reached experimental validation, which we use as the comparison class throughout this paper. We claim novelty for none of these.

That compilation also sets the scale against which this campaign must be judged. Across the 56 entries carrying a numeric hit rate, **239 of 459 synthesised compounds were active at 10 micromolar or better, a pooled hit rate of 52 percent**, with a median of 5 compounds synthesised per campaign. Generative design routinely reaches the bench and routinely works. Any account of the failure reported here has to explain why this campaign did not, against 73 that did.

What this paper contributes is a chain with **ground truth at the end of it**. The molecules did not stop at a benchmark. Three went to synthetic chemists, and the synthesis did not succeed. That allows a question the literature rarely gets to ask: when a computational campaign fails at the bench, was the information needed to prevent it absent, or present and unused?

For this campaign the answer is unambiguous and is the paper's central finding. The information was present, in the campaign's own spreadsheet, in a column named after the functional group that broke the chemistry.

We are the authors of the generative system audited here (ref. 13). This is a self-audit, and we regard that as a strength of the evidence rather than a limitation of it: we hold the primary artefacts, including the failure.

2. Provenance of the material audited

```
Medgnosis GenAI system  (published, ref. 13)
             |
             v
KRAS campaign: 73 molecules generated
             |
             v
Docking: DiffDock + GNINA, 4 KRAS variants
             |
             v
Property prediction: ADMET-AI
PAINS | AMES | QED | SA score | Epoxide Ring Present
             |
             |  ....... computed, recorded,
             |          NOT wired to any decision
             v
SELECTION RULE:  DiffDock conf >= -1.5
                 GNINA affinity <= -5.0 kcal/mol
             |
             v
Shortlist recorded (73 molecules + 2 reference drugs)
             |
             v
3 molecules sent to synthetic chemists
             |
             v
```

```
~1 year, no compound delivered
```

Figure 1. The chain, and where it breaks. Property prediction (D) fed the record but not the decision (E). The dotted edge is the paper's subject.

The chain from published method to bench failure is summarised in Figure 1. The generative system and its selection rule are published (ref. 13): molecules were retained on a DiffDock confidence threshold of −1.5 and a GNINA minimised affinity of −5.0 kcal/mol, with ADMET-AI (ref. 12) descriptors computed alongside. No stability or reactivity criterion appears in the method. The campaign output analysed here is the spreadsheet released with this paper, containing 73 generated molecules and two reference drugs, sotorasib (ref. 9) and adagrasib (ref. 10).

3. Results

3.1 The failure was recorded in the file that recommended the molecules

The campaign's output file carries 32 columns. Among them is `Epoxide Ring Present`.

Table 1. What the campaign computed about the molecules it selected.

quantity recorded	value across 73 generated molecules	reference drugs
`Epoxide Ring Present`, the three sent to synthesis	**True, all three**	not applicable
`Epoxide Ring Present`, all molecules	True for 13 of 73	False
`AMES` (predicted mutagenicity), median	**0.93**	not applicable
`QED` (drug-likeness), median	0.39	typical norm > 0.5
`Synthesis Accessibility Score`, median	5.47 (22 of 73 above 6.0)	**4.21, 3.84**
`PAINS`	flagged 4 of 73	not flagged
`Good Docking Quality Overall`	**True for 72 of 73**	not applicable
`FDA Approved`	False for 73 of 73	not applicable

Three molecules were selected for synthesis: Hit 13, Hit 41 and Hit 73 (Figure 2). Each has `Epoxide Ring Present = True`. Epoxides are strained three-membered ethers, opened by nucleophiles, acids and bases, and in these structures they co-occur with hydrazines and free aldehydes that attack them. The audit script identifies twelve molecules whose own functional groups react with one another, including all three synthesis candidates.

The signal was not weak, ambiguous or buried. It was a boolean column named after the offending group, set to True, in the file used to choose what to make.

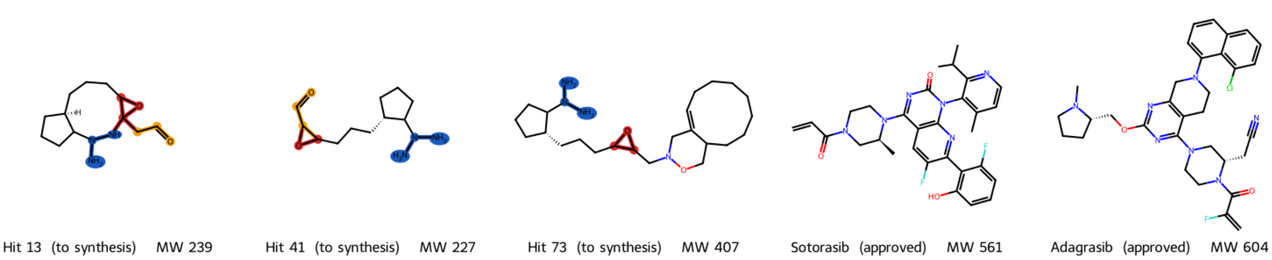

Figure 2. The three molecules sent to synthetic chemists, beside the two reference drugs in the same file. Red, epoxide. Blue, hydrazine or triazane nitrogen. Orange, free aldehyde. Structures drawn from the SMILES strings in Table 2; no atom has been moved or omitted.

Table 2. The three synthesis candidates as recorded in the campaign output. SMILES strings are listed verbatim beneath the table.

molecule	MW	SA score	aromatic ring	groups present
Hit 13	239	5.24	**no**	epoxide, hydrazine, triazane, aldehyde
Hit 41	227	4.60	**no**	epoxide, hydrazine, triazane, aldehyde
Hit 73	407	5.29	**no**	epoxide, hydrazine, triazane
Sotorasib (ref. 9)	561	3.84	yes	none of the above
Adagrasib (ref. 10)	604	4.21	yes	none of the above

SMILES as recorded in the campaign output file:

```
Hit 13   NN(N1)C2CCC[C@H1]2CCCC3OC31CC=O
Hit 41   NN(N)C1CCC[C@H1]1CCCC2OC2C=O
Hit 73   NN(N)C1CCC[C@H1]1CCCC2OC2CN(OCC3CCCCCCCC=4)CC=43
```

Each of the three carries **an epoxide and a hydrazine in the same molecule**. A hydrazine is a strong nucleophile and an epoxide is a strained electrophile; they consume one another. Two of the three additionally carry a free aldehyde, which condenses with the hydrazine to a hydrazone. Every one contains a triazane, a nitrogen-nitrogen-nitrogen chain that is notoriously unstable.

The two approved drugs sit in the same spreadsheet, carry none of these groups, contain aromatic rings, and are roughly twice the molecular weight. The comparison was available at the moment of selection.

3.2 The failure is not a property of generative models

If generative models generally produced unmakeable molecules, this campaign would be an instance of a known problem rather than a finding. We tested that directly against the released outputs of eight published generative models, applying **BRENK and PAINS exactly as shipped in RDKit**, with none of our own modifications.

Table 3. One filter battery, unmodified published catalogues, applied across models.

set	n	pass filter	aromatic ring	median MW
Purchasable compounds (ZINC)	30,000	**97.5%**	100.0%	523.6
HMM	3,381	89.4%	77.6%	108.1
AAE	27,730	87.7%	97.9%	323.4
VAE	29,297	86.5%	98.3%	306.4
CharRNN	29,653	86.2%	98.2%	310.4
JT-VAE	30,000	86.0%	98.6%	307.8
NGram	7,134	85.5%	89.9%	233.3
LatentGAN	26,799	84.7%	97.5%	307.4
Combinatorial	30,000	81.5%	95.4%	325.4
This campaign	**73**	**6.8%**	**0.0%**	226.3

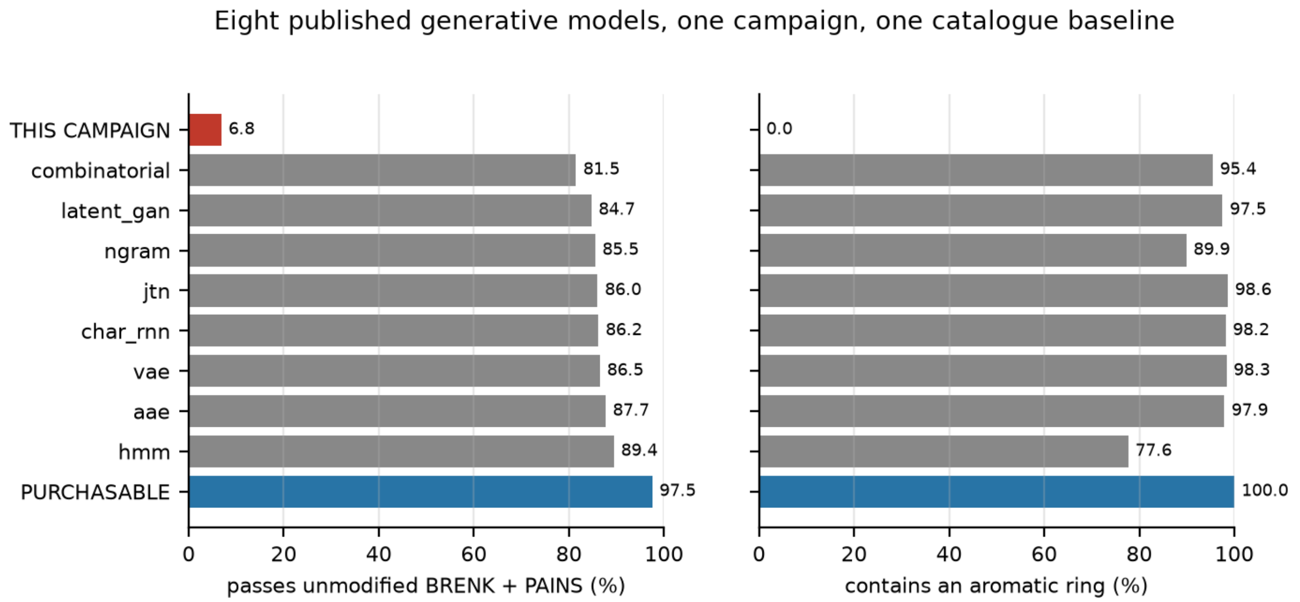

Figure 3. One filter battery across nine sources. Left, fraction passing unmodified BRENK and PAINS as shipped in RDKit (ref. 21). Right, fraction containing an aromatic ring. Red, the audited campaign. Blue, purchasable compounds. Grey, eight published generative models whose released outputs are distributed with the MOSES benchmark (ref. 15).

Eight published models, 240,214 molecules, pass at 81 to 89 percent (Figure 3). This campaign passes at 6.8 percent, and **not one of its 73 molecules contains an aromatic ring**, against 95 to 99 percent for every model tested and 100 percent for purchasable compounds. Both reference drugs contain aromatic rings.

The generative literature does not have this problem. This campaign did. The difference lies in what was permitted to stop a molecule, not in what produced it.

A methodological note on this table. The filters are unmodified. Had we applied our own tuned catalogue, the separation would be uninterpretable, because we would have built the instrument that produced it.

3.3 The same unmodified filter rejects both approved drugs

Applying that identical unmodified catalogue to the two reference drugs:

Table 4. The filter that separates the sets also rejects the drugs.

molecule	status	passes unmodified BRENK + PAINS
Sotorasib	approved KRAS G12C inhibitor (ref. 9)	**No**
Adagrasib	approved KRAS G12C inhibitor (ref. 10)	**No**

Both fail, on the acrylamide warhead that BRENK flags as `Michael_acceptor_1` (ref. 5). The alert fires on the feature responsible for the drug's mechanism.

This is the second half of the problem and it points the opposite way. Used as a gate without calibration, the same catalogue that correctly rejects this campaign's molecules also rejects the two drugs the campaign was trying to emulate. A filter is not a gate until something has established which of its verdicts to believe.

3.4 Three mechanisms of silent failure, five instances

Re-running each component of the vetting stack against molecules whose correct classification is known yields five failures. They reduce to three mechanisms.

Table 5. Failures grouped by mechanism.

mechanism	instance	consequence
M1. Over-broad structural alert	BRENK `Michael_acceptor_1` on acrylamide	rejects sotorasib and adagrasib
	Nitroso SMARTS `[NX3]-[OX2H0,OX1]` also matches nitro	rejects venetoclax
	Alkyne rule in our own stability gate	rejects MRTX1133 (ref. 7)
M2. Threshold not fitted to the chemotype	ADMET DILI < 0.70 and hERG < 0.70 (ref. 12)	rejects 199 of 200, including both approved drugs
M3. Tool answers a different question	Retrosynthesis (ref. 11) as a feasibility gate	ranks an impossible molecule above a marketed drug

We report three mechanisms rather than five failures deliberately. Grouping is the honest description, and the recurrence of M1 across three independent implementations, two of them widely used and one written by us, is stronger evidence than five unrelated defects would be.

For M3, the numbers: retrosynthesis returns **80 percent of precursors in stock over 6 steps** for an impossible molecule ("Hit 41", which carries both an epoxide and a hydrazine that opens it) and **71 percent over 6 steps** for adagrasib. A formal disconnection exists for a molecule that decomposes on formation. Retrosynthesis answers "if this existed, what would assemble it" and never "can this exist".

In every instance the tool returned well-formed output and raised no error.

3.5 The intervention: panels, not references

All five instances are detected by requiring each tool to reproduce known answers before its output is used. The operative variable is panel size.

Table 6. Detection as a function of control-panel size.

control set	size	instances detected
Reference compounds	3	**0 of 5**
Calibration panel	14	**5 of 5**

The panel spans the target class, approved covalent inhibitors, and structurally diverse marketed drugs. Cost is one function call per tool invocation. In the rebuilt pipeline the ADMET stage re-runs its control check on every invocation and exits if any control fails.

3.6 The rebuilt pipeline, and what its gates cost

Eleven phases, each with a known-answer gate. **Nine of the eleven gates exist because a tool had already failed silently.**

Table 7. Method validation across four targets.

variant	structure	cognate redock RMSD	control spread	criterion
G12C	8AFB	0.67 Å	2.96 kcal/mol	< 2.0 Å
G12D	9HFK	0.88 Å	4.84 kcal/mol	< 2.0 Å
G12V	9YMQ	1.55 Å	not recorded	< 2.0 Å
G12R	9XB7 (1.36 Å)	0.99 Å	3.13 kcal/mol	< 2.0 Å

The method replaced was blind docking with CNN rescoring, which showed **4.0 kcal/mol run-to-run variance on a single molecule** (sotorasib scored −3.80 and −7.78 on repeat runs) and therefore could not rank.

Target selection by disease prevalence. Among 3,755 PDAC tumours carrying a codon-12 KRAS mutation, the distribution is **G12D 47%, G12V 34%, G12R 17%, G12C 2%** (ref. 8). *(Denominator: codon-12-mutant tumours. As fractions of all PDAC, approximately 40, 29, 14 and 1.7 percent.)* The audited campaign's docking columns lead with **G12C**, the variant with approved drugs and abundant liganded structures, and the one accounting for 2 percent of this disease. Structure availability selected the target; prevalence did not.

Screens. From 518,662 gated purchasable compounds: G12D n = 19,639, G12V n = 9,913, covering 81 percent of codon-12-mutant PDAC. Neither produced a compound competitive with the clinical inhibitors. A G12R screen is in progress.

3.7 Our replacement methods failed in the same way, and the gates caught them

Two failures occurred in the pipeline built to prevent failures. Both were detected by control gates, which is the intervention demonstrating itself on its authors.

A ligand scored outside the protein. Our MM-GBSA implementation initially scored a conformer re-embedded from SMILES rather than the docked pose, placing ligand and receptor in unrelated coordinate frames. The **crystallographic** cognate ligand of 9YMQ scored **+23.44 kcal/mol**, non-binding, and impossible for a ligand resolved within that structure. Scoring the docked pose gives **−33.22 kcal/mol**. The gate that caught it required the cognate ligand to score as a binder.

A gate that did not check what it claimed. An earlier version of that same gate counted returned results rather than testing the ranking its documentation promised, and allowed the broken run to proceed for 37 minutes. A gate that does not check what it claims to check is indistinguishable from no gate. This is the paper's thesis, applied to the authors, and it is the reason we report it.

Rescoring cannot rank a shortlist flat in the primary score.

Table 8. Two independent rescoring methods on the G12V top-200.

quantity	value
Vina score spread across all 200 compounds	**1.80 kcal/mol**
Vina's documented error (ref. 6)	~2.5 kcal/mol
MM-GBSA spread, same compounds	87.28 kcal/mol
Spearman , Vina vs MM-GBSA (n = 200)	**+0.106** (p = 0.13)
Spearman , Vina vs Vinardo (n = 200)	+0.259
Shared compounds in the two top-10 rankings	**2 of 10**

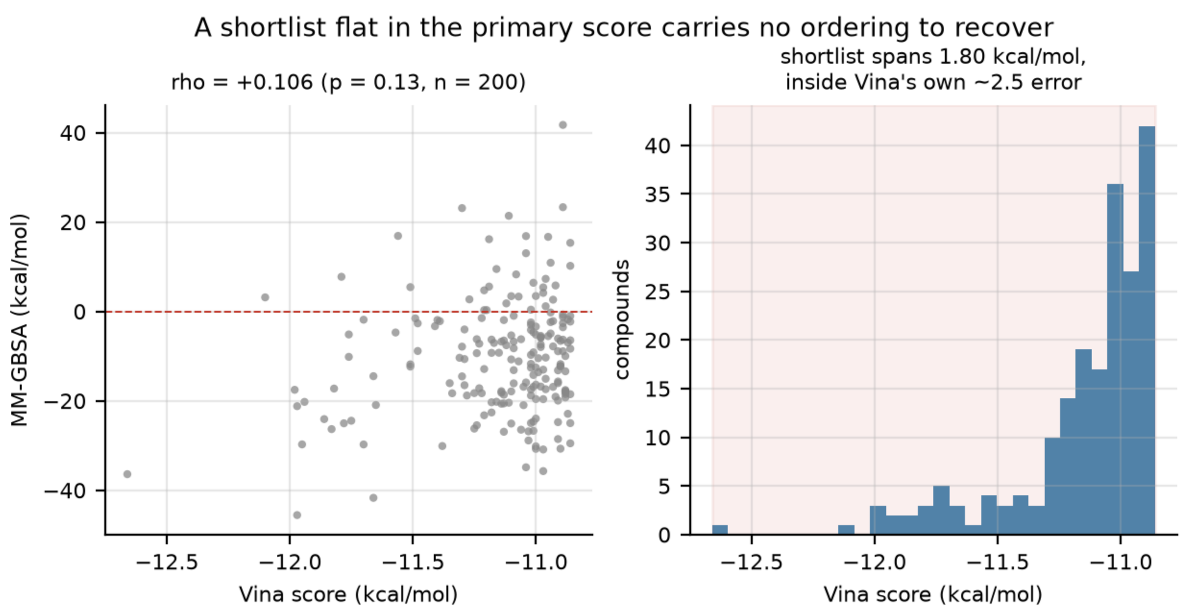

Figure 4. A flat shortlist carries no ordering to recover. Left, MM-GBSA against Vina for all 200 G12V compounds. Right, the Vina score distribution, spanning 1.80 kcal/mol against a documented method error of about 2.5 kcal/mol.

Neither rescoring method is malfunctioning (Figure 4). The shortlist spans less than the primary method's own error and carries no ordering to recover. Reported without the primary spread, = 0.106 reads as a failed rescoring method rather than a property of the input. Interim estimates during the run were 0.239 (n = 50), 0.347 (n = 60), 0.196 (n = 80) and 0.106 (n = 200); the apparent significance at n = 60 ($p = 0.007$) was a sampling artefact, and we report it to document how unstable small-n correlations are in this setting.

Search effort buys geometry, not ranking. Every screen ran at docking exhaustiveness 4 while every validation gate ran at 16, so the gate never covered the configuration used.

Table 9. Exhaustiveness sensitivity, 50 compounds, identical embedding seed.

exhaustiveness	mean Δ vs ex128	vs ex128	top-10 retained	median cost
4	**+0.08 kcal/mol**	0.821	8 of 10	24 s
16	+0.02	0.950	9 of 10	94 s
64	−0.01	0.977	10 of 10	347 s
128	0.00	1.000	10 of 10	747 s

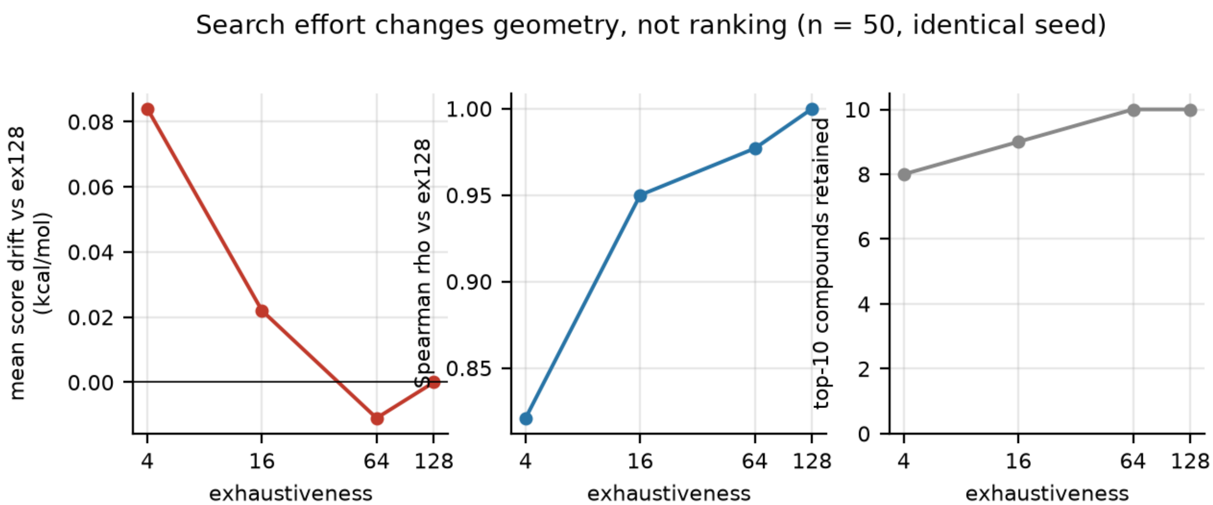

Figure 5. Search effort changes geometry, not ranking. Fifty compounds docked at four exhaustiveness levels with an identical embedding seed, so the only variable is the search. Left, mean score drift against the deepest search. Centre, rank correlation. Right, top-10 compounds retained.

A 31-fold compute increase moves the median score by 0.08 kcal/mol (Figure 5), so the completed screens are sound. The same parameter on G12R moved the cognate's score only from −9.28 to −9.80 while moving its **pose from 7.19 Å to 0.99 Å**, converting a failed validation into a passed one. Screening consumes rank and tolerates shallow search; validation and pose-dependent rescoring consume geometry and do not.

3.8 Selecting hard on docking does not, by itself, reduce makeability

If a docking objective pulled a pipeline away from real chemistry, the effect should be visible inside a purchasable library. We tested this on 1,986 G12D-screened purchasable compounds, binned into deciles by Vina score.

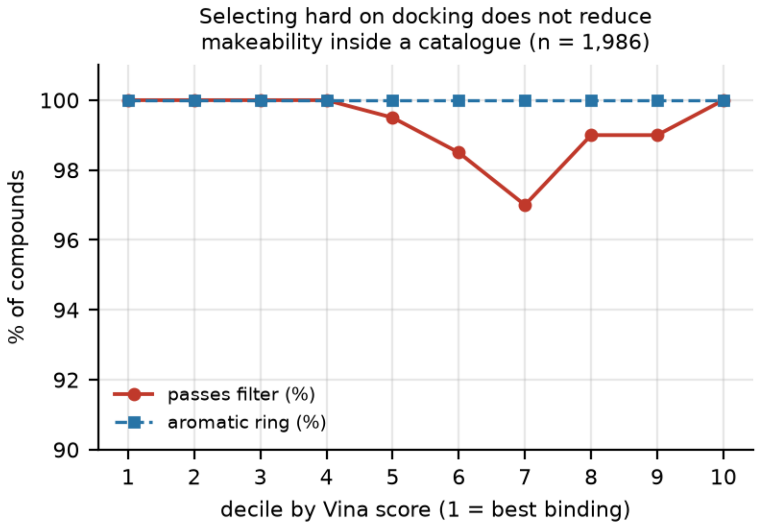

Figure 6. Docking rank against makeability, inside a catalogue. Unmodified filter pass rate and aromatic content by Vina decile, best binding on the left.

Table 10. Makeability is flat across the docking range.

decile	Vina range (kcal/mol)	passes filter	aromatic	median MW
1 (best)	−12.50 to −10.01	100.0%	100%	520.5
2	−10.00 to −9.57	100.0%	100%	516.2
5	−9.03 to −8.76	99.5%	100%	517.6
7	−8.45 to −8.15	97.0%	100%	516.7
10 (worst)	−7.03 to −4.51	100.0%	100%	533.6

There is no gradient (Figure 6). Selecting hard on a docking objective is safe when the search space is a catalogue, because the catalogue bounds the search to chemistry that exists. The failure documented here therefore required **both** an unconstrained generative search space **and** the absence of any criterion able to reject a molecule. Neither alone reproduces it.

We report this because it refutes a mechanism we ourselves proposed on the strength of section 3.2, and because it is the control a reader is entitled to demand.

3.9 Gate coverage does not predict whether molecules get made

The obvious explanation for the campaign's failure is that too few of its computed properties were wired to decisions. We defined **gate coverage** as the fraction of computed property types carrying a threshold able to remove a molecule, and tested whether it separates successful campaigns from this one.

Papers were drawn as a seeded random sample from the curated corpus of 73 generative campaigns with experimental validation compiled by Du et al. (ref. 14). Scoring followed a rubric fixed before any paper was read (released with this manuscript). Four of twelve sampled papers were retrievable as open-access full text and were scored.

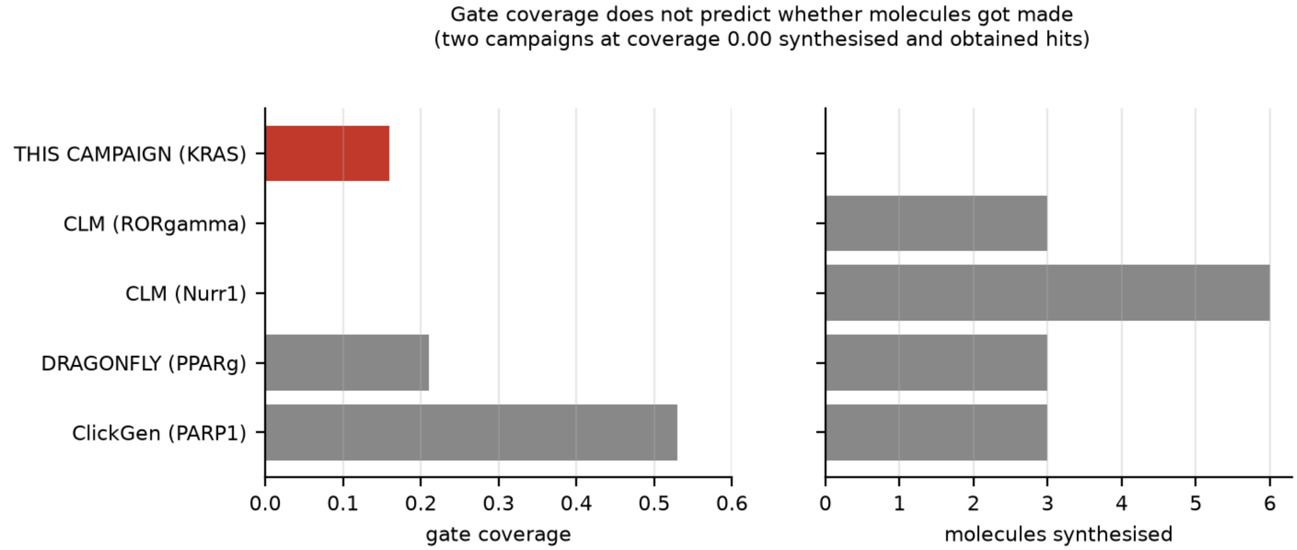

Figure 7. Gate coverage against outcome. Left, fraction of computed properties able to halt the pipeline. Right, molecules actually synthesised. Red, the audited campaign.

Table 11. Pilot survey of gate coverage (n = 4 scored, rubric fixed in advance).

campaign	ref.	computed	gating	coverage	stability gate	synthesised	hits
ClickGen (PARP1)	17	15	8	**0.53**	**yes**	3	2
DRAGONFLY (PPAR-gamma)	18	14	3	0.21	no	3	2
CLM (Nurr1)	19	8	0	**0.00**	no	6	2
CLM (ROR-gamma)	20	4	0	**0.00**	no	3	**3**
This campaign	13	19	3	0.16	no	**0**	0

The hypothesis fails (Figure 7). Two campaigns with gate coverage of zero synthesised their designs and obtained hits, one of them three from three at 0.37 micromolar (ref. 20). Automated gate coverage does not distinguish them from a campaign that produced nothing.

What the ROR-gamma paper does record is a different kind of gate. Its first round of designs was judged "synthetically inaccessible by medicinal chemists", retrosynthetic analysis "did not find a synthetic route", and that assessment prompted a revision of the method rather than an order for synthesis (ref. 20). The gate existed, it was a person, and it fired before money was spent. The campaign audited here has no equivalent step in its record: the spreadsheet in section 3.1 was produced and three molecules were requested from chemists.

The claim that survives is narrower than the one we set out to make, and is about process rather than tooling. **A criterion capable of stopping the pipeline must exist and must be consulted. It does not have to be automated, and automating it is not sufficient.**

3.10 Three hypotheses of ours, tested and refuted

We state these together because the pattern is the paper's most useful methodological content.

Table 12. Hypotheses proposed during this work and their outcomes.

hypothesis	test	outcome
Generative models produce molecules that cannot be made	filter battery, 240,214 molecules, 8 models (3.2)	**refuted**, they pass at 81 to 89 percent
Goal-directed optimisation degrades makeability	decile analysis (3.8), plus 50 goal-directed campaigns pooling a 50 percent hit rate (ref. 14)	**refuted**
Low gate coverage explains the failure	pilot survey, 4 campaigns (3.9)	**refuted**, two campaigns at coverage 0.00 succeeded

Each was plausible, each would have made a more quotable paper, and each was removed by a control we ran on ourselves. What remains is a single documented negative case with an unusually complete audit trail, set against 73 positive controls from the literature.

3.11 The 2025 campaign and the rebuilt pipeline, stage by stage

Table 13 sets the audited campaign beside its replacement. The 2025 column is taken from the published method (ref. 13) and from the column headings of the output file itself; it is not a characterisation. The last two columns record only one thing about each stage: whether it could remove a molecule. **GATE** means it could, **rank** means it ordered molecules without removing any, and **none** means the stage produced a record only.

Table 13. Campaign as run in 2025 against the rebuilt pipeline. *The 2025 column is drawn from a published method; the rebuilt column is our own account of our own unpublished system. The evidential standards are not equal and the comparison should be read accordingly.*

stage	2025 campaign	rebuilt pipeline	2025	now
Target choice	G12C leads the docking columns	chosen by PDAC prevalence (ref. 8); cognate ligand required	none	**GATE**
Method validation	none reported	cognate redock RMSD < 2.0 A	none	**GATE**
Scoring reliability	not assessed	4.0 kcal/mol variance measured, method replaced	none	**GATE**
Docking	DiffDock blind, GNINA rescoring	site-directed Vina 1.2.7 (ref. 6)	rank	rank
Library	generated, unconstrained	518,662 purchasable, MW-windowed	none	**GATE**
Chemical reality	epoxide and PAINS flags computed	14-drug calibration panel must survive	none	**GATE**
Selection heuristic	not applicable	similarity plus diversity, validated vs random	none	rank
Selection rule	DiffDock >= -1.5, GNINA <= -5.0	docking rank after all gates pass	**GATE**	**GATE**
ADMET	AMES, ClinTox, QED, Lipinski computed	thresholds from approved controls; exits on failure	none	**GATE**
Synthetic accessibility	SA score computed, median 5.47	retrosynthesis plus chemoselectivity check	none	**GATE**

stage	2025 campaign	rebuilt pipeline	2025	now
Independent rescoring	none	MM-GBSA, gated on the cognate scoring as a binder	none	**GATE**
Laboratory capture	none	specified, **not built**	none	none
Stages able to stop a molecule			**1 of 12**	**9 of 12**

Two features of this table matter more than the rest.

Most of the 2025 stages existed. The campaign computed reactive-group flags, PAINS, mutagenicity and synthetic accessibility. In seven of the twelve rows the difference is not whether a property was calculated but whether the calculation could remove a molecule. One stage in the 2025 campaign could stop a molecule; nine can now.

The rebuilt pipeline has not produced a compound either. It has produced four validated targets, three completed screens and a set of gates, and no purchase order. On the only measure that finally matters, a molecule in a vial with a number attached, the two columns read the same. The difference claimed here is in what is known about why, not in a better outcome.

4. Discussion

The components of this stack behaved as documented. Structural alerts over-rejected, as Capuzzi et al. described (ref. 2). Retrosynthesis did not model stability, which it never claimed to. Docking scores were noisy within their stated error. Nothing malfunctioned.

What failed was the architecture. Property prediction produced a record; selection consulted a threshold. Between the column reporting an epoxide and the decision to synthesise the molecule carrying it, there was no connection. The campaign could see, and could not act on what it saw.

This distinction matters because it changes the remedy. If the problem were prediction quality, the response would be better predictors, and the field is well supplied with those. If the problem is that predictions are recorded rather than enforced, better predictors change nothing. The remedy is architectural: every stage must be able to stop the pipeline, and must first demonstrate on known answers that it stops the right things.

The benchmark in §3.2 supports this reading. Eight published models produce broadly makeable molecules without any special stability machinery, because they imitate training distributions drawn from real chemistry. This campaign selected against docking thresholds and departed from that distribution, with nothing positioned to notice. **Not one of its 73 molecules has an aromatic ring**, a feature present in essentially every drug-like compound and in both its own reference drugs. A single aromaticity check would have flagged the entire set.

The pilot survey (section 3.9) constrains this further than we would like. Two campaigns with no automated gates at all synthesised their molecules and obtained hits. The mechanism cannot therefore be gate coverage as a number. What distinguishes the successful campaigns in the sample is that a criterion, in one documented case a medicinal chemist's judgement, was positioned to reject designs and did so before synthesis was commissioned. The architectural claim survives; its automated form does not.

We report our own two failures for the same reason we report the campaign's. A paper arguing that computational tools fail silently would be self-refuting if it presented its own tools as having worked first time. The relevant difference is that ours were caught, by gates, before anything was ordered.

5. Limitations

One campaign, no control campaign. We cannot separate "the vetting architecture failed" from "this particular campaign was unusual". The benchmark in §3.2 bounds the second possibility but does not eliminate it.

The synthesis record is not in our possession. The approximately one-year effort is reported from institutional memory. No laboratory notebook, dated protocol or chemist's report accompanies this paper. Readers should treat the bench outcome as attested rather than documented, and the computational audit, which is fully reproducible, as the evidential core.

n = 73. The audited set is small. All per-molecule claims are stated with counts.

The 14-drug panel is sufficient, not optimal. It detects these five instances. Minimal sufficient panel construction is open.

One target family. All docking results are KRAS.

The gate-coverage survey is a pilot, n = 4. Twelve papers were sampled at random from the corpus; eight were not retrievable as open-access full text. Four scored papers cannot establish a population claim, and the survey is reported as refuting our hypothesis rather than establishing its replacement. A full survey requires institutional access and is the obvious next step.

Three of our own hypotheses were refuted during this work (section 3.10). We report them rather than the tidier account that would have followed from stopping earlier, because the refutations bound what the surviving claim can carry.

Self-audit. We built the system audited here. This gives us the primary artefacts and an obvious interest in the interpretation. All data and code are released so the analysis can be repeated against us.

6. Methods

Software. AutoDock Vina 1.2.7 (ref. 6); RDKit 2026.03 (BRENK and PAINS catalogues as shipped); OpenMM 8.6 with openmmforcefields, openff-toolkit and AmberTools (AM1-BCC charges via antechamber), ff14SB with GBn2 implicit solvent; AiZynthFinder with USPTO policy and ZINC stock (ref. 11); ADMET-AI (ref. 12); Meeko for ligand preparation; PDBFixer for receptor preparation.

Structures. 8AFB (G12C), 9HFK (G12D), 9YMQ (G12V), 9XB7 (G12R, 1.36 Å). Intake required a co-crystallised drug-like ligand. Docking boxes were the ligand extent plus 10 Å per dimension.

Filter battery (§3.2). BRENK and PAINS as shipped in RDKit, no exemptions. Generated sets are the released samples of the eight MOSES baseline models, capped at 30,000 molecules each. Molecules failing SMILES parsing are excluded and counted.

Gates. Cognate redock RMSD < 2.0 Å; survival of a 14-drug panel; approved-control retention in ADMET; stability gate before retrosynthesis; cognate ligand scoring as a binder with sensible control ordering in MM-GBSA.

Statistics. Spearman rank correlation throughout, reported with n. Interim estimates are reported alongside final values where they differ materially.

7. Data and code availability

`github.com/eobi/pancreatic_cancer_research` contains code, gated libraries, screening results, rescoring outputs, the filter battery, and all validation records **including failed validations**. The G12R validation that failed at 7.19 Å is retained alongside the one that passed at 0.99 Å. A record containing only successes cannot support a methodological claim.

8. References

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Supplementary reference list: gate-coverage survey corpus

The papers below are the contemporary campaigns (2024 to 2026) retrieved as the sampling frame for the gate-coverage survey in section 3.9. They are listed so the frame is inspectable, and are not cited individually in the argument.

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